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Problem 2 - Eigenvalues Depending on a Free Parameter

Tag List: [STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC]

Consider the LTI system

$$\begin{cases} \dot{x}(t) &= A x(t) + B u(t) \\ y(t) &= C x(t) + D u(t) \end{cases}$$

where the matrices are respectively:

$$A = \begin{bmatrix} 0 & 1 \\ -10(3+4k) & -(13+2k) \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ +1 \end{bmatrix} \quad C = [40k \quad +2k] \quad D = [0]$$

and k is a strictly positive parameter: $k \in \mathbb{R}, k > 0$

Assignment:

- Determine the expression of the eigenvalues λ_1, λ_2 of the matrix A .
- Compute the eigenvalues when the gain k assumes the following values: $k = 0.1, 1, 100$
- Determine the characteristic polynomial $p_A(s, k)$ and of the matrix A .
- Compute the values of the parameter k that lead to coincident real eigenvalues. Once the values of k have been found, calculate the corresponding eigenvalues.
- Determine the value of the parameter k that leads to have the eigenvalues $\lambda_{1,2} = -20 \pm j \sqrt{170}$.

Your code shall

- Define the **symbolic parameter** k and the **symbolic** complex **variable** s .
- Define the **symbolic matrix** A as in the above expression.
- Determine the expression of the eigenvalues of A . Sort the eigenvalues in ascending order and store those expressions in the **symbolic column vector** $\mathbf{eigA}$.
- Evaluate the expression of the eigenvalues of A when $k = 0.1, 1, 100$. Store the results in the **symbolic array** $\mathbf{eigSET}$. Each column of $\mathbf{eigSET}$ must correspond to a different value of the parameter k . Moreover, each column of $\mathbf{eigSET}$ has two elements.
- Determine the expression of the characteristic polynomial of the matrix A as a **symbolic function** of the complex variable s and the parameter k : $\mathbf{pA(s, k)}$.
- Compute the values of the parameter k that leads to real coincident eigenvalues for the matrix A .
- **Hint:** the characteristic polynomial $p_A(s, k)$ is a second-order polynomial, so to have two real and coincident roots, the **discriminant** of such a polynomial must be ...
- Sort the resulting values of k and store them in the symbolic column vector $\mathbf{k_sol}$.
- Once the values of $\mathbf{k_sol}$ have been found, calculate the corresponding eigenvalues, sort them and store them in the symbolic array $\mathbf{eig_sol}$. Each column of $\mathbf{eig_sol}$ corresponds to a different $\mathbf{k_sol}$ value.
- Determine the value of the parameter k that leads to the eigenvalues $\bar{\lambda}_{1,2} = -20 \pm j \sqrt{170}$.
- **Hint:** if $\bar{\lambda}_{1,2}$ are the actual eigenvalues, then a value $\hat{k}$ must exist, such that $p_A(\bar{\lambda}_1, \hat{k}) = 0$ and $p_A(\bar{\lambda}_2, \hat{k}) = 0$. Solving one of those equations allow us to find the desired value for k .

- Store the solution in the symbolic variable **k_d**.
- As validation of the finding, compute the eigenvalues of the matrix A when the parameter k assumes the value **k_d**, sort the eigenvalues and save them in the symbolic array **eigA_kd**.

Solution Proposal

```
syms s
syms k positive
syms A [2,2]
syms B [2, 1]
syms C [1, 2]

A = [0, 1; -10*(3+4*k), -(13+2*k)];
B = sym([0;1]);
C = [40*k, 2*k];

eigA = sort(eig(A),'ascend')
```

eigA =

$$\left(\begin{array}{c} -k-\frac{\sqrt{4\,k^2-108\,k+49}}{2}-\frac{13}{2} \\ \frac{\sqrt{4\,k^2-108\,k+49}}{2}-k-\frac{13}{2} \end{array}\right)$$

```
kVALS = [0.1, 1, 100];
eigSET = sym(zeros(2, numel(kVALS)));

for n=1:numel(kVALS)
    dummyVAL = subs(eigA, k, kVALS(n));
    eigSET(:, n) = dummyVAL;
end
% eigSET = sort(eigSET, 'ascend');
disp(eigSET)
```

$$\left(\begin{array}{ccc} -\frac{\sqrt{956}}{10}-\frac{33}{5} & -\frac{15}{2}-\frac{\sqrt{55}\,i}{2} & -\frac{\sqrt{29249}}{2}-\frac{213}{2} \\ \frac{\sqrt{956}}{10}-\frac{33}{5} & -\frac{15}{2}+\frac{\sqrt{55}\,i}{2} & \frac{\sqrt{29249}}{2}-\frac{213}{2} \end{array}\right)$$

```
pA(s, k) = charpoly(A,s)
```

$$pA(s, k) = s^2 + (2\,k + 13)\,s + 40\,k + 30$$

```
discriminant(k) = (2*k+13)^2-4*1*(40*k+30);
k_sol = sort(solve(discriminant==0), 'ascend')
```

k_sol =

$$\begin{pmatrix} \frac{27}{2} - \sqrt{170} \\ \sqrt{170} + \frac{27}{2} \end{pmatrix}$$

```
k1_sol = k_sol(1); k2_sol = k_sol(2);
eigA1 = simplify(subs(eigA,k, k1_sol))
```

$$\text{eigA1} = \begin{pmatrix} \sqrt{170} - 20 \\ \sqrt{170} - 20 \end{pmatrix}$$

```
eigA2 = simplify(subs(eigA, k, k2_sol))
```

$$\text{eigA2} = \begin{pmatrix} -\sqrt{170} - 20 \\ -\sqrt{170} - 20 \end{pmatrix}$$

```
eig_sol = [eigA1, eigA2]
```

$$\text{eig_sol} = \begin{pmatrix} \sqrt{170} - 20 & -\sqrt{170} - 20 \\ \sqrt{170} - 20 & -\sqrt{170} - 20 \end{pmatrix}$$

```
eig_d = sym([-20-1i*sqrt(170); -20+1i*sqrt(170)]);
```

```
pA_kd(k) = collect(simplify(pA(eig_d(1),k)), k)
```

$$pA_kd(k) = (-2\sqrt{170}i)k + 27\sqrt{170}i$$

```
k_d = solve(pA_kd==0)
```

$$k_d = \frac{27}{2}$$

```
eigA_kd = sort(simplify(subs(eigA,k, k_d)), 'ascend')
```

$$\text{eigA_kd} = \begin{pmatrix} -20 - \sqrt{170}i \\ -20 + \sqrt{170}i \end{pmatrix}$$

